

THE UNIVERSITY OF AZAD JAMMU & KASHMIR, MUZAFFARABAD



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DEPARTMENT OF SOFTWARE ENGINEERING

SD&A

Lab 8

Objective

The objective of this lab was to design low-fidelity wireframes for a "University Student Portal" Android application using Balsamiq. The purpose of this activity was to visualize the application's layout, navigation flow, and user interface components before the actual development phase. The design focuses on usability, clarity, and ease of navigation for a student user base.

Application Overview

The **University Student Portal App** is a mobile platform designed to digitize academic management for students. It serves as a central hub where students can perform essential tasks such as registering for courses, checking attendance, viewing results, and managing their financial status (fees).

Key Features:

- **Secure Authentication:** A simple login interface with password recovery options.
- **Centralized Dashboard:** A grid-based home screen for quick access to all modules.
- **Academic Management:** interfaces for course registration and dropping subjects.
- **Progress Tracking:** Visual representation of attendance records.
- **Profile Management:** Settings to update personal contact information and security credentials.

Wireframe Screen Descriptions

Figure 1: Login Screen

The login screen is designed with a minimalist approach to reduce cognitive load. It features a prominent "Student Portal" header and an academic icon to establish identity. The input fields for "Username" and "Password" are centered and widened for easy thumb-reach on mobile devices. A primary "Login" button uses a distinct color to draw attention, while a "Forgot Password" link provides a recovery route.

Figure 2: Dashboard

The Dashboard serves as the main navigation hub. The top section displays the user's profile summary (Name: Haseeb, Reg No: 1554) to confirm identity. The navigation is structured as a **2x2 Grid**, utilizing large, labeled buttons with icons (Courses, Attendance, Results, Fee). This layout ensures that the most frequently used features are immediately accessible without scrolling.

Figure 3: Available Courses

This screen allows students to manage their enrollments. It utilizes a list view layout. Each row displays the Course Code, Course Name, and a clear "Action" link (Register or Drop). The use of a data grid structure keeps the information organized and scannable. A back button is included in the top header for easy navigation back to the Dashboard.

Figure 4: Attendance Screen

The Attendance screen prioritizes data visualization. Instead of raw text tables, **progress bars** are used to indicate attendance percentages for each subject (e.g., Operating Systems, Data Structures). This allows students to instantly gauge their standing (e.g., identifying low attendance risks) at a glance.

Figure 5: Profile & Settings

This screen allows users to view and edit their personal details. Standard text input fields are provided for Name, Email, and Phone Number. To prevent accidental changes, specific "Change Password" and "Save Profile" buttons are placed at the bottom. The layout adheres to standard mobile form design patterns, with labels top-aligned for readability.

Usability Principles Applied

In designing these wireframes, several core **Heuristic Evaluation** principles were applied:

1. **Visibility of System Status:** The Attendance screen uses visual progress bars to give immediate feedback on the student's status, rather than forcing them to calculate percentages manually.
2. **Consistency and Standards:** The application maintains a consistent layout across all screens. The top header (Time, Battery, Signal) and the "Back" navigation arrow are placed in the same location on every sub-screen, ensuring a predictable user experience.
3. **Aesthetic and Minimalist Design:** Clutter is minimized by removing non-essential elements. The Login and Dashboard screens only present the necessary options, keeping the interface clean and focusing the user's attention on the primary tasks.
4. **Recognition Rather Than Recall:** The Dashboard uses intuitive icons (a Book for Courses, a Calendar for Attendance) alongside text labels. This helps users recognize the function of a button visually without having to read every label intently.

5. Conclusion

This lab successfully demonstrated the importance of low-fidelity prototyping. By using Balsamiq, I was able to iterate on the layout, specifically improving the spacing of input fields and the organization of the dashboard grid, without writing code. The resulting wireframes provide a clear blueprint for the development team, ensuring that the final Android application will be user-friendly and structurally sound.



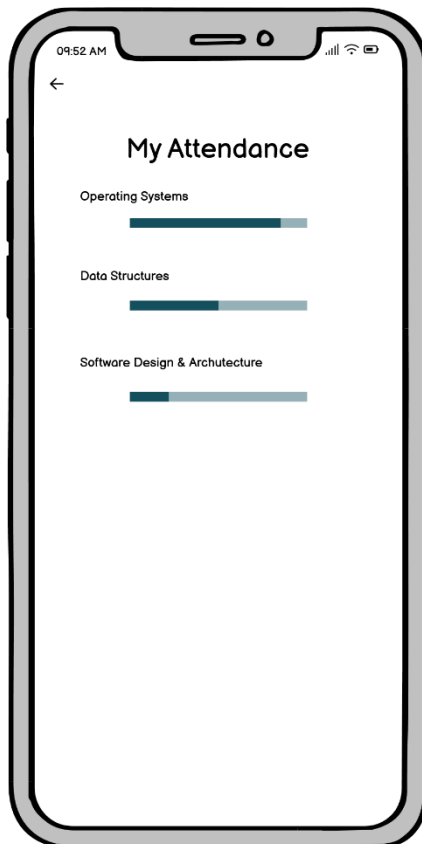
Login



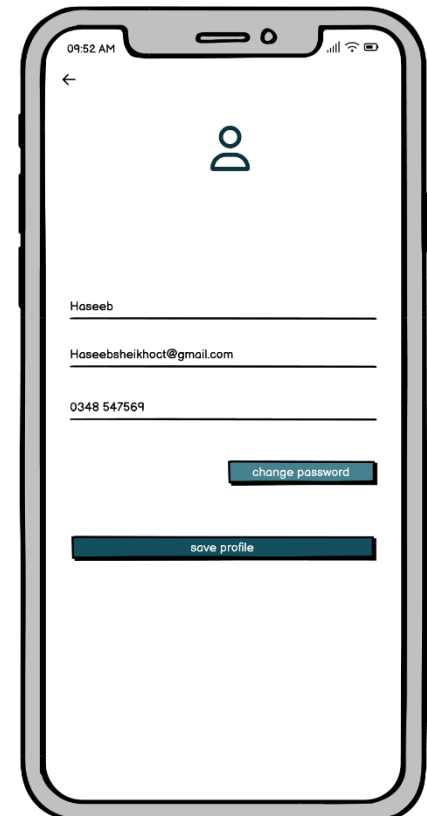
Dashboard



Courses



Attendance



Settings